

SAARTHI AI

A Smart Guardian App for Women's Safety using AI & ML

> “When danger whispers, Ssarthi listens, watches, and protects.”

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Problems Faced by Women in Today's World:

Fear of being followed, harassed, or attacked

— especially when alone at night

Inability to call for help

during panic or physical danger

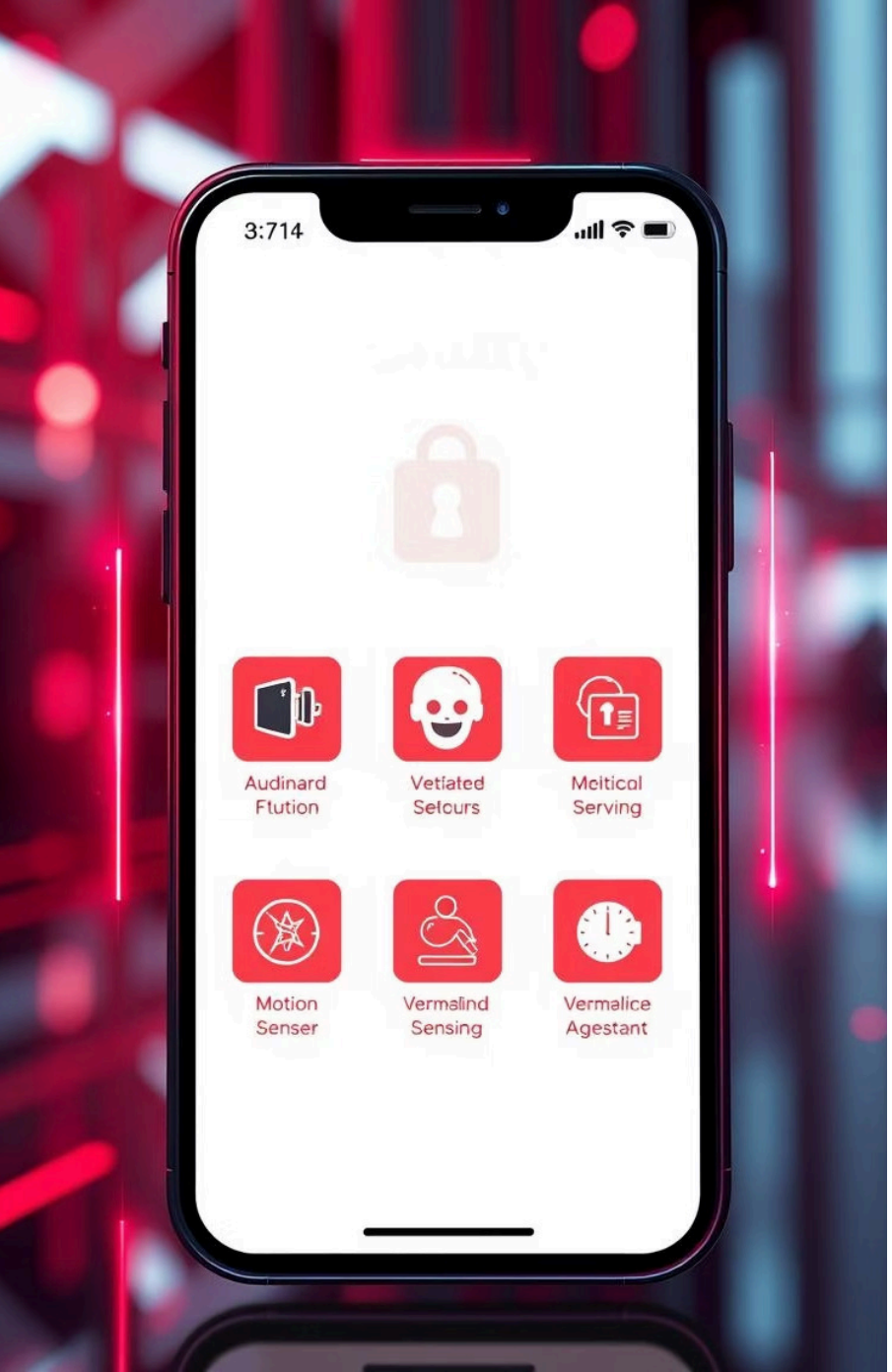
Existing safety apps require manual actions

like pressing SOS, which isn't always possible

Lack of prediction-based safety

— current tools are reactive, not proactive

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Problem Solved by the AI/ML Model:

Using Artificial Intelligence & Machine Learning, Saarthi AI can:

- **Detect early signs of danger from surroundings**
  using audio, camera, and motion sensors
- **Predict possible sexual assault intent**
  through facial expressions, voice tone, and abnormal movement
- **Auto-trigger alerts, location sharing, and navigation to safe areas**
  — without manual input
- **Learn from user behavior**
  to become smarter and more responsive over time

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Steps of Implementation & Use of Code:

1. Dataset Collection & Preprocessing

→ Facial emotion (FER-2013), audio threats (RAVDESS, CREMA-D), motion patterns (HAR/WISDM)

2. Model Training

→ CNN for facial expression detection

→ LSTM or MFCC + CNN for audio abuse detection

→ Anomaly detection using accelerometer + gyroscope data

3. On-Device Conversion

→ Convert models to TensorFlow Lite (.tflite) for mobile usage

4. App Development

→ Android Studio or Flutter

→ Use Google Maps API, Firebase, Twilio for alerting

5. Integration

→ Models run in background using sensors

→ If threat detected: vibrate + show warning + send location to contacts + map rerouting

6. Testing & Deployment

→ Real-time testing in controlled environments

→ Final packaging for demo / Play Store

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Future Advancements with AI:

- AI that adapts to the user's unique fear triggers & locations
- Integration with smart wearables (bracelets, rings) for physical triggers
- Real-time crowd-sourced danger zone maps
- Automatic video streaming to police or guardians
- Offline mode with edge-AI (no internet required)
- Support for children, elderly, and travelers

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Scenario – When Saarthi Speaks Before She Does

Scenario:

Ananya, a 20-year-old college student, is walking home after a late class.

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She scrolls her phone with earphones in, unaware of her surroundings.

Suddenly...

A man starts following her — steps get faster, closer.

He mutters something inappropriate under his breath.

Her facial expression shifts from calm to confused... to tense.

🧠 Saarthi AI springs into action — without being touched:

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Detects the threatening voice behind her

Identifies Ananya's rising anxiety through micro-expressions

Notices an unusual walking pattern from motion sensors

Then, Ananya softly says:

🗣️ **"Saarthi, help."**

In seconds, Saarthi responds:

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✅ **Vibrates strongly and flashes: “Threat Detected Nearby”**

✅ **Sends her live GPS + 10-second voice clip to her parents**

✅ **Opens Google Maps and reroutes her to the nearest police station**

✅ **Displays the “Emergency Exit” button to call 112 directly**

⏱️ No delay. No panic. Just action.

Ananya follows the path. The man backs off. She reaches safety.

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